

Activity: Poisson assumptions and diagnostics

Model

```
m_art <- glm(art ~ phd + ment + kid5,  
             family = poisson, data = articles)
```

Characteristic	IRR	95% CI	p-value
phd	1.01	0.96, 1.06	0.8
ment	1.03	1.02, 1.03	<0.001
kid5	0.91	0.84, 0.97	0.005

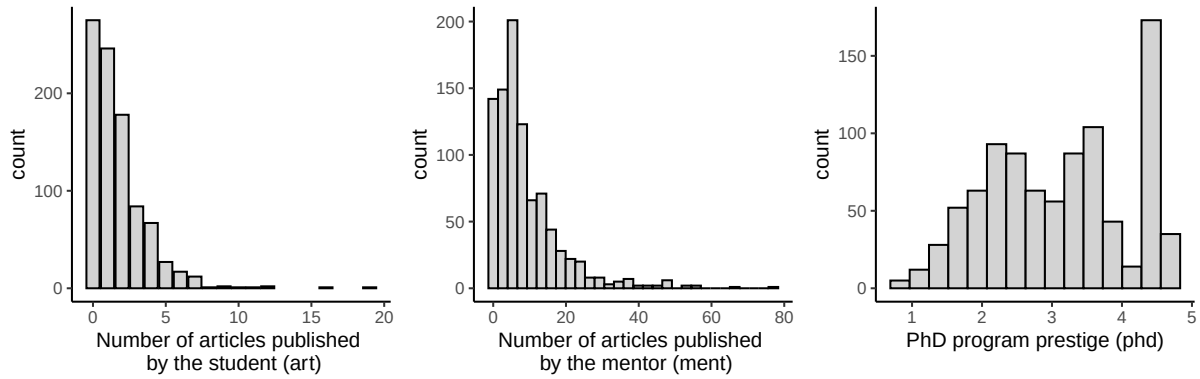
Abbreviations: CI = Confidence Interval, IRR = Incidence Rate Ratio

(The *Incidence Rate Ratio* is just e^β , i.e. the multiplicative change in the estimated mean response).

Data and EDA

Characteristic	N = 915 ¹
art	1 (0, 2)
phd	3.15 (2.26, 3.92)
ment	6 (3, 12)
kid5	
0	599 (65%)
1	195 (21%)
2	105 (11%)
3	16 (1.7%)

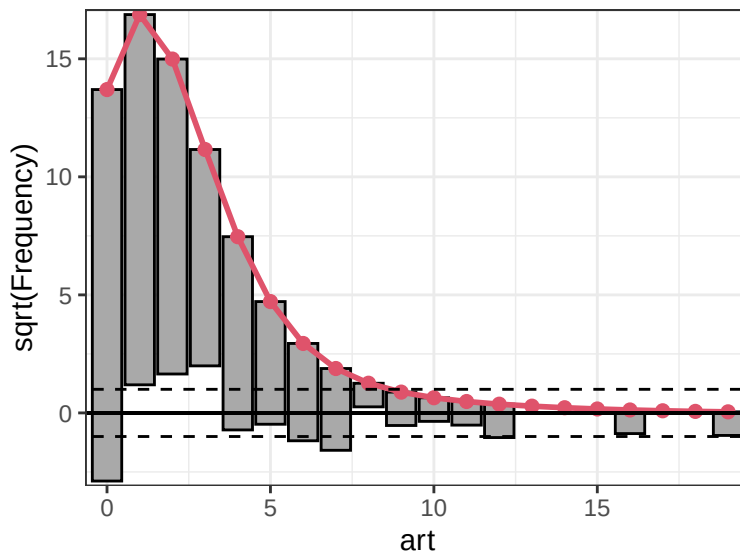
¹Median (Q1, Q3); n (%)



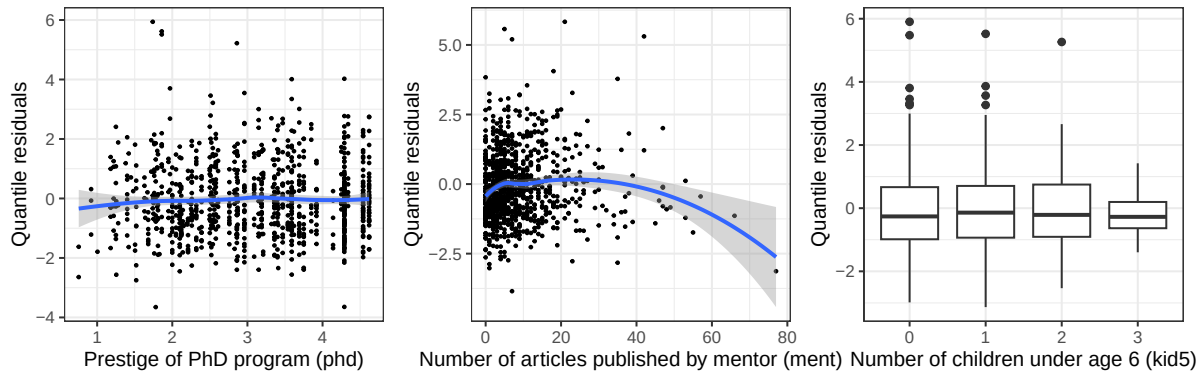
Rootogram

A *rootogram* is a tool used to assess overall model fit, by assessing whether the predicted counts from the fitted model match the observed counts from the data.

- horizontal axis: possible values for the response variable
- vertical axis (square root scale): predicted count for the model (points) and *observed* counts from the data (bars)
- Lack of fit could be due to:
 - Wrong choice of distribution
 - Additional explanatory variables need to be added to model
 - Issues with shape assumption



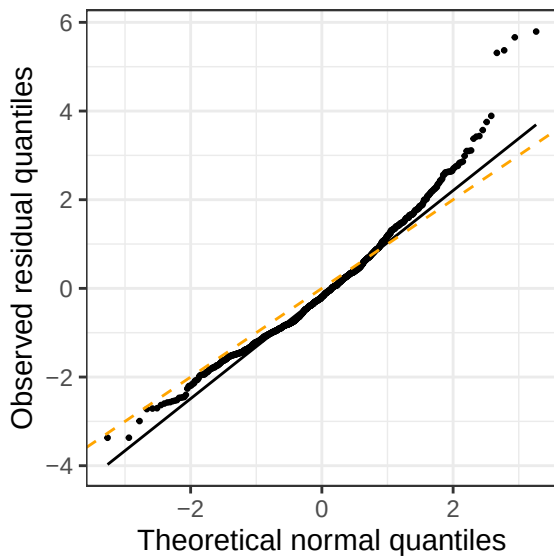
(Randomized) quantile residual plots



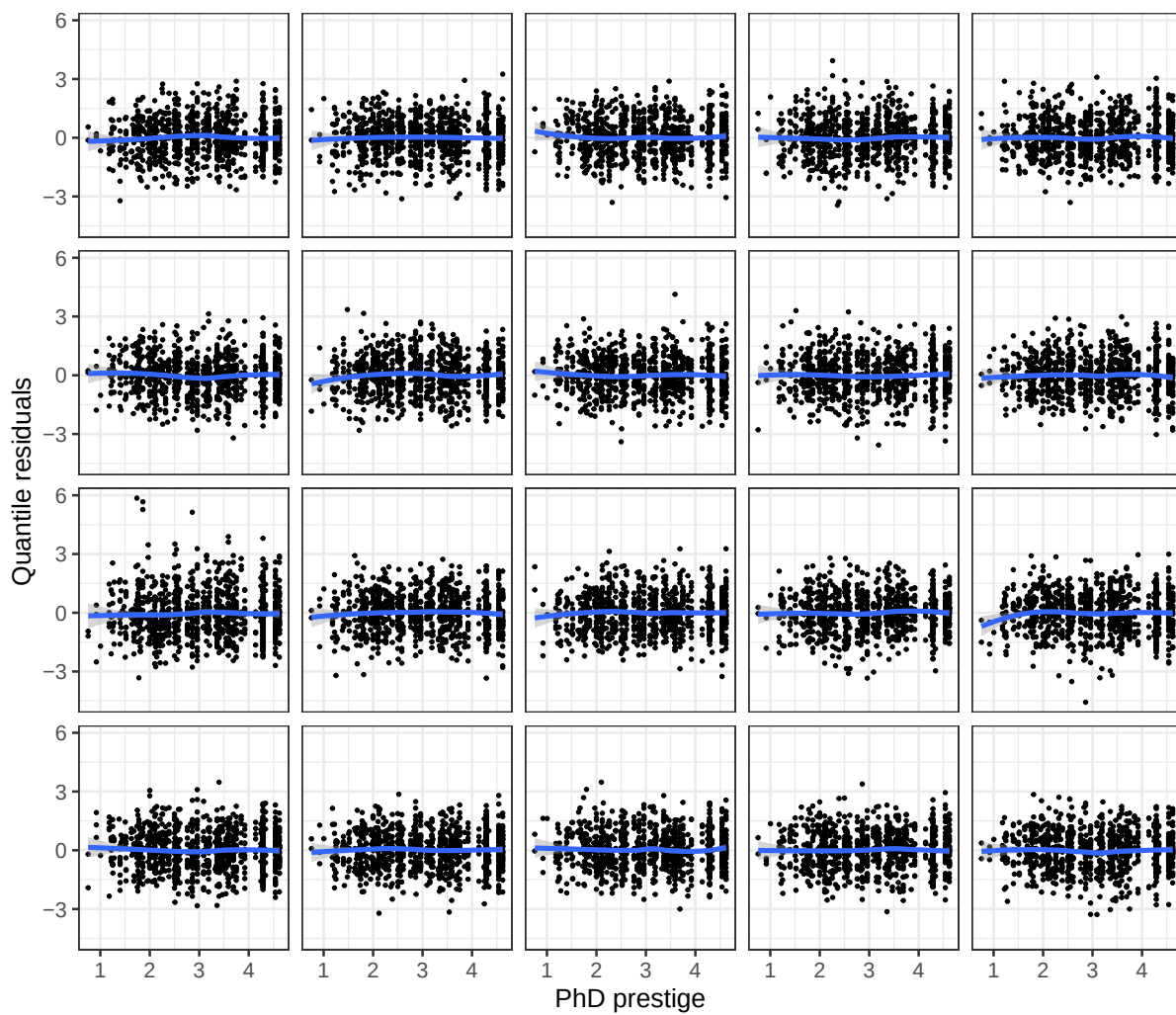
QQ plot of quantile residuals

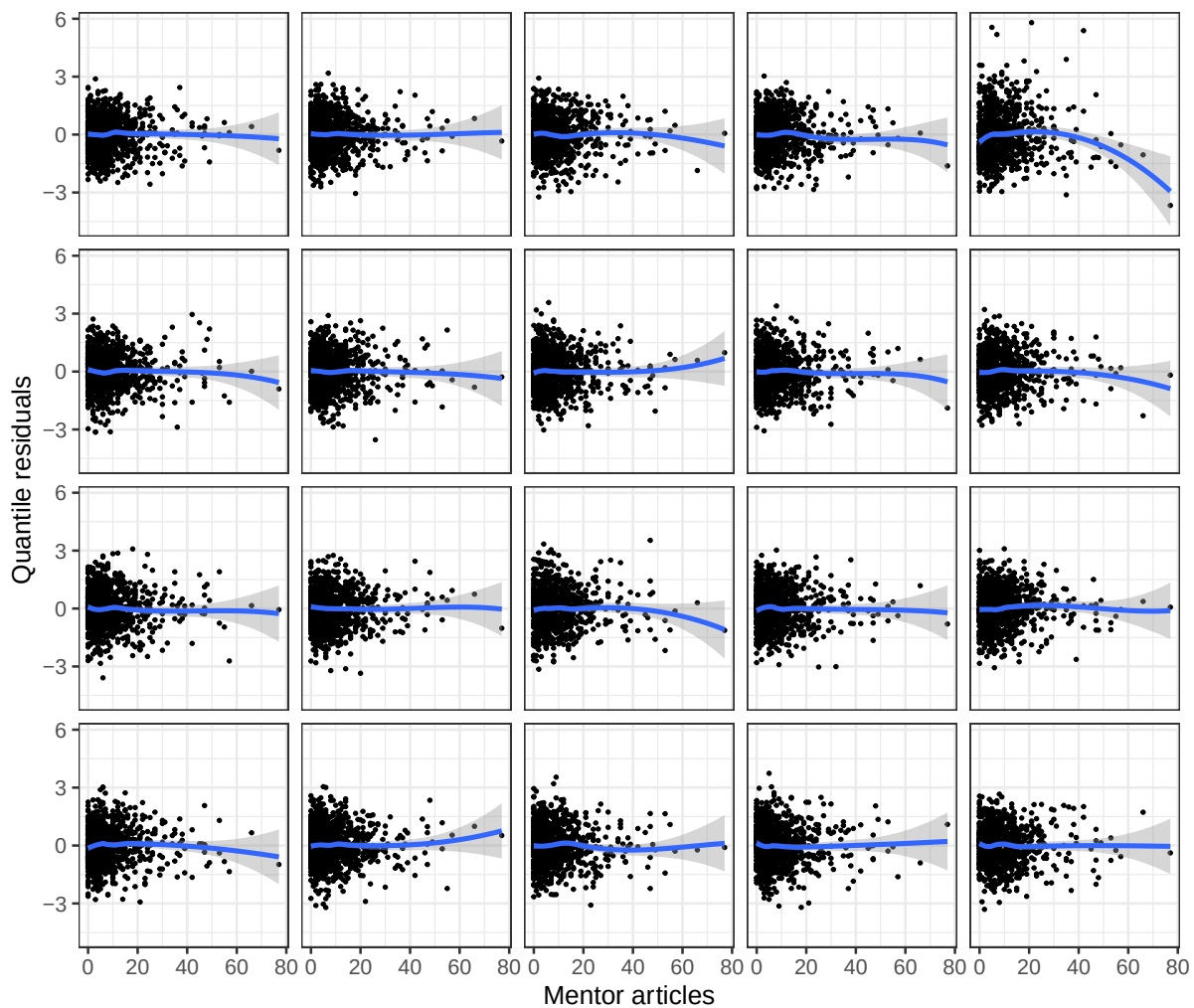
If the model is a good fit to the data, randomized quantile residuals should be approximately $N(0, 1)$.

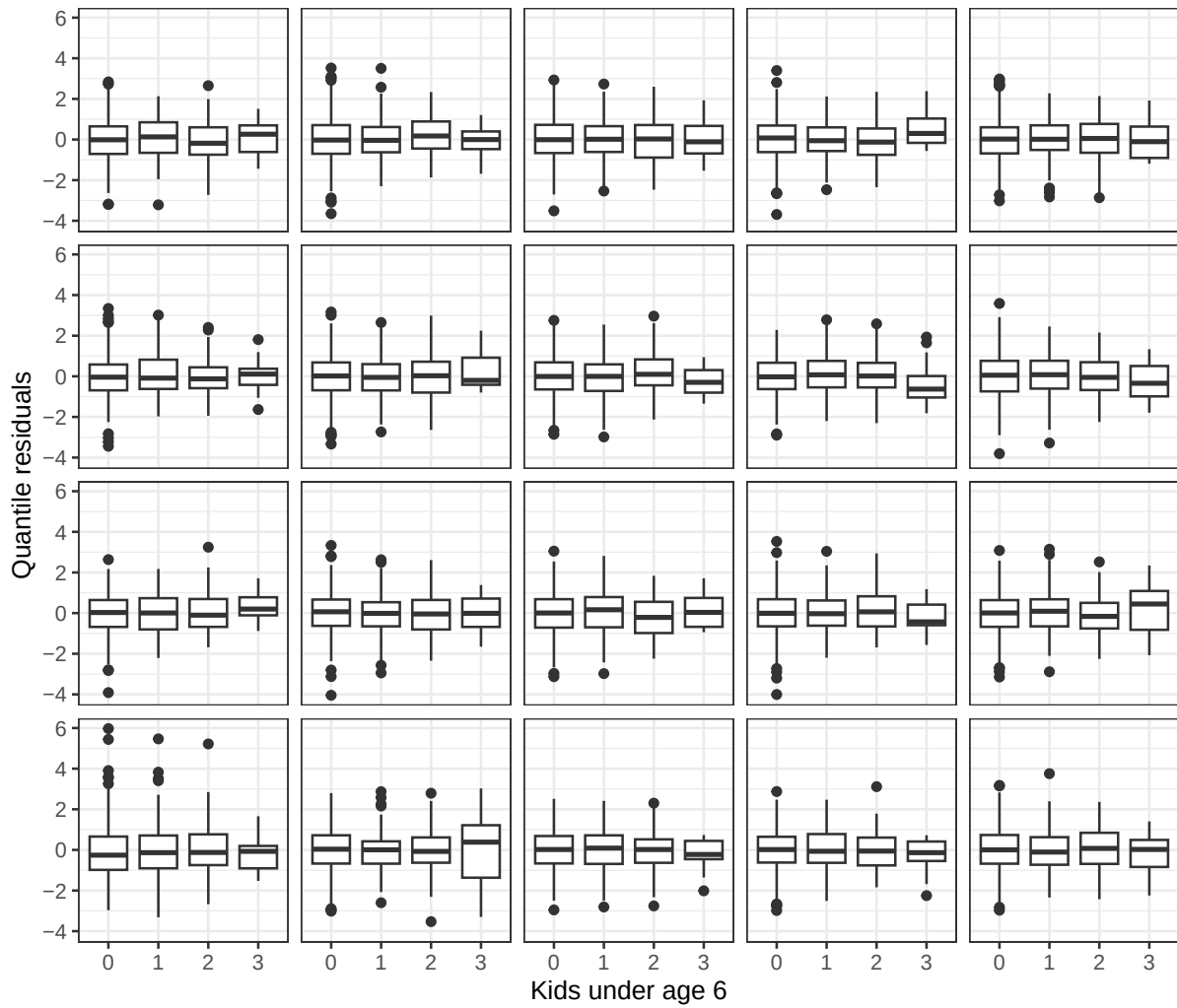
The solid black line in the plot is drawn through the 1st and 3rd quartiles of the residuals (this is how a line on a QQ plot is usually created). The dashed orange line is the line of slope 1 and intercept 0.



Plot lineups







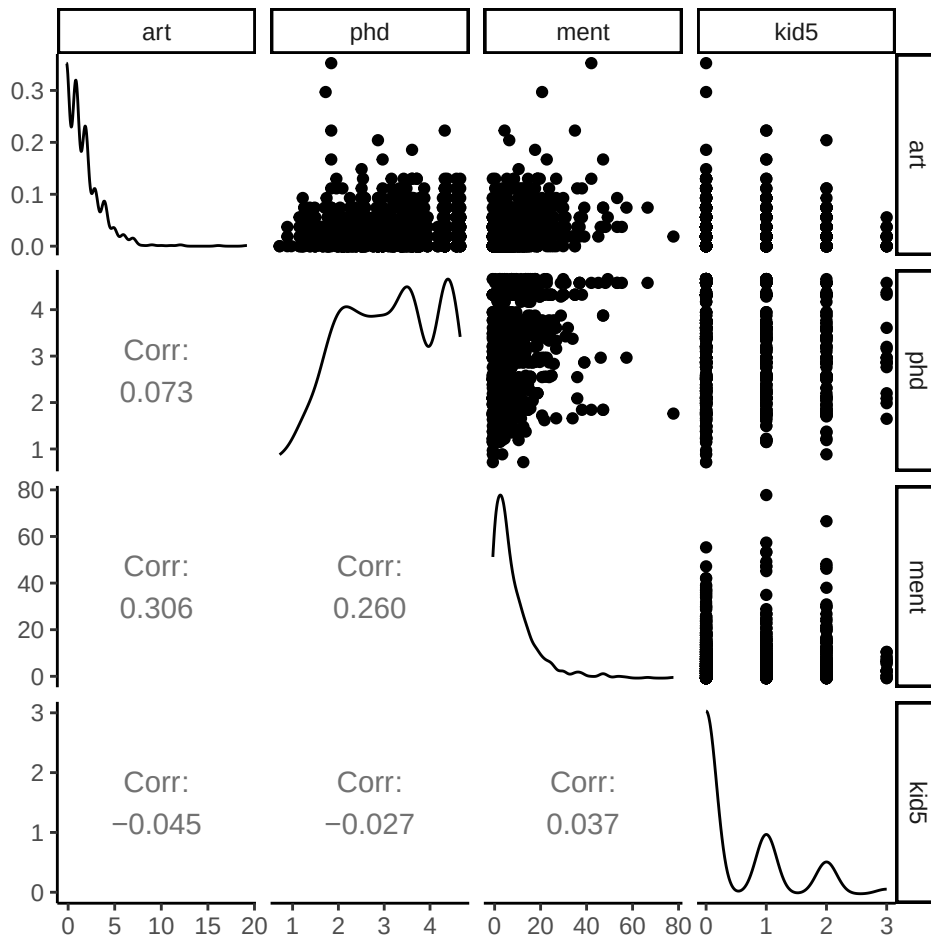
Cook's distance

The following table shows the first few rows of the data with their associated Cook's distances, ordered by descending Cook's distance. (The numbers next to the table are just the row numbers in the original data).

	art	phd	ment	kid5	cooks_distance
328	1	1.78	77	1	0.5587585
915	19	1.86	42	0	0.4113048
914	16	1.74	21	0	0.1478051
911	11	2.86	7	2	0.0905633
912	12	4.29	35	1	0.0638557

	art	phd	ment	kid5	cooks_distance
913	12	1.86	5	1	0.0625560
909	9	1.86	47	1	0.0481382
669	2	4.54	55	0	0.0351612
803	4	4.54	66	2	0.0243971
883	6	2.51	11	2	0.0199222

Correlation and VIFs



```
car::vif(m_art)
```

```
      phd      ment      kid5
1.057959 1.071738 1.015631
```